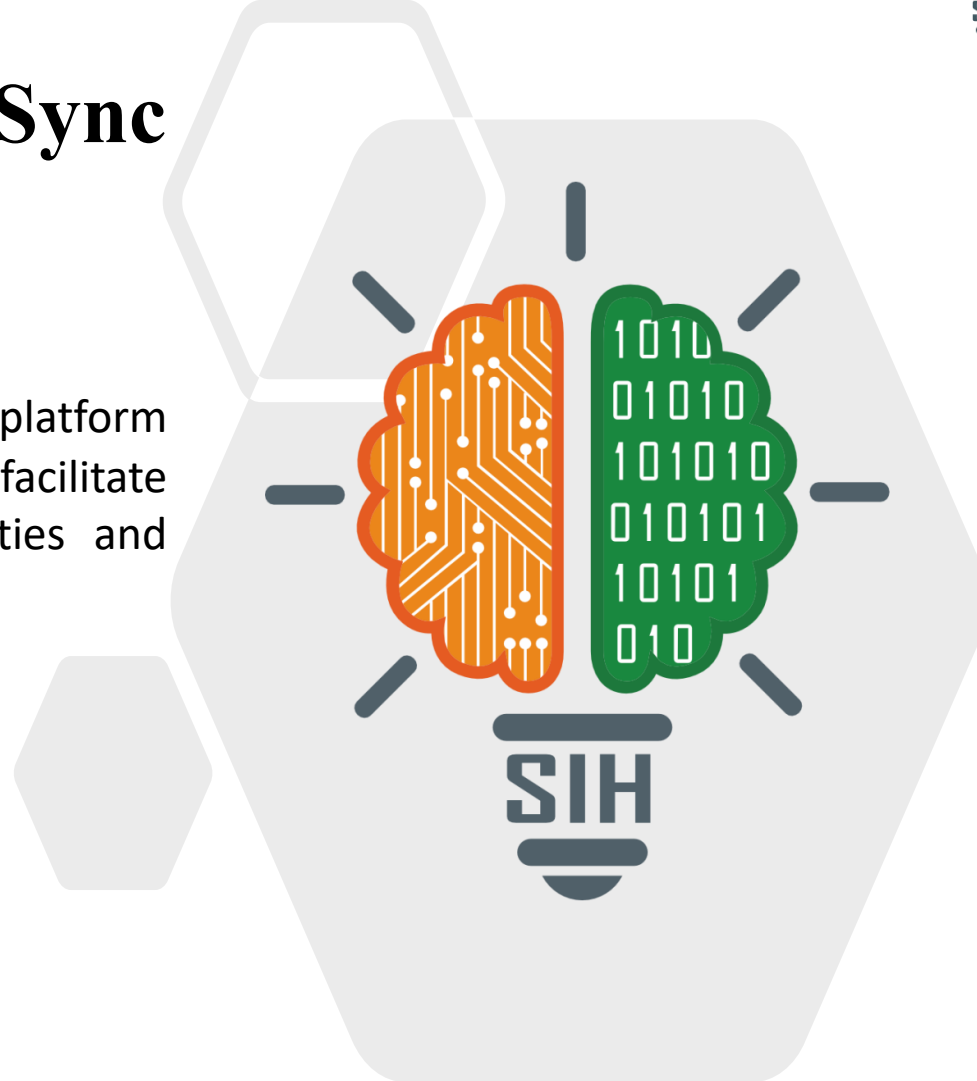




CivicSync

- **Problem Statement ID – 26043**
- **Problem Statement Title-** A digital platform to crowdsource societal challenges and facilitate collaborative problem solving through universities and industry partnerships
- **Theme- Smart Education**
- **PS Category- Software**
- **Team ID-**
- **Team Name- Mind Mavericks**



Problem

❖ **Problems remain unstructured & fragmented**

Citizen complaints come from multiple channels, but similar issues are rarely clustered, prioritized, and converted into clear, actionable problems.

❖ **Expertise remains disconnected**

Universities, faculty, students, researchers, and industry have relevant expertise, but the right problem-solving team is difficult to identify and connect.

❖ **The solution journey is disconnected**

Problem discovery, solution development, implementation, and impact measurement often happen separately — creating a gap from **Problem** → **Expertise** → **Solution** → **Implementation** → **Impact**.

Proposed Solution

➤ **End-to-End Collaborative Problem Solving**

CivicSync connects challenges with **students, researchers, universities & industry** to develop, implement, and **track real-world impact**.

Problem → **Expertise** → **Solution** → **Impact**

➤ **AI-Powered Challenge Intelligence**

CivicSync converts citizen reports into **structured, prioritized challenges**, identifies required expertise, and **clusters related issues**.

Understand → **Cluster** → **Prioritize**

Innovation & Uniqueness

1. AI Problem
Intelligence

Converts citizen
reports into
structured
challenges

2. Problem
Clustering &
Priority

Detects duplicate
issues & prioritizes
by impact

3. Explainable
Capability
Matching

Matches problems
with students,
universities &
industry

4. Solution-to-
Implementation

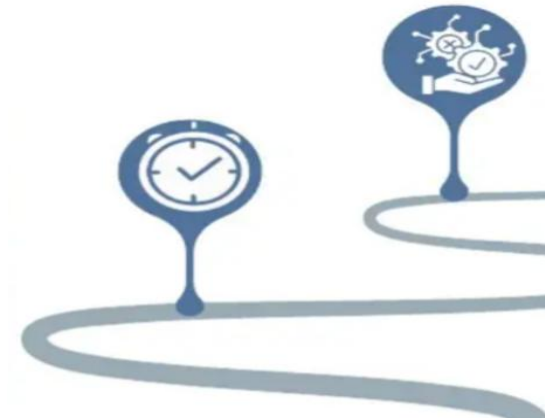
Connects validated
solutions with
startups & MSMEs

5. Evidence-
Backed Impact

Tracks Baseline to
Target to Actual
with verification

6. Impact
Passport &
Reputation

Transparent Impact
Passport & verified
reputation



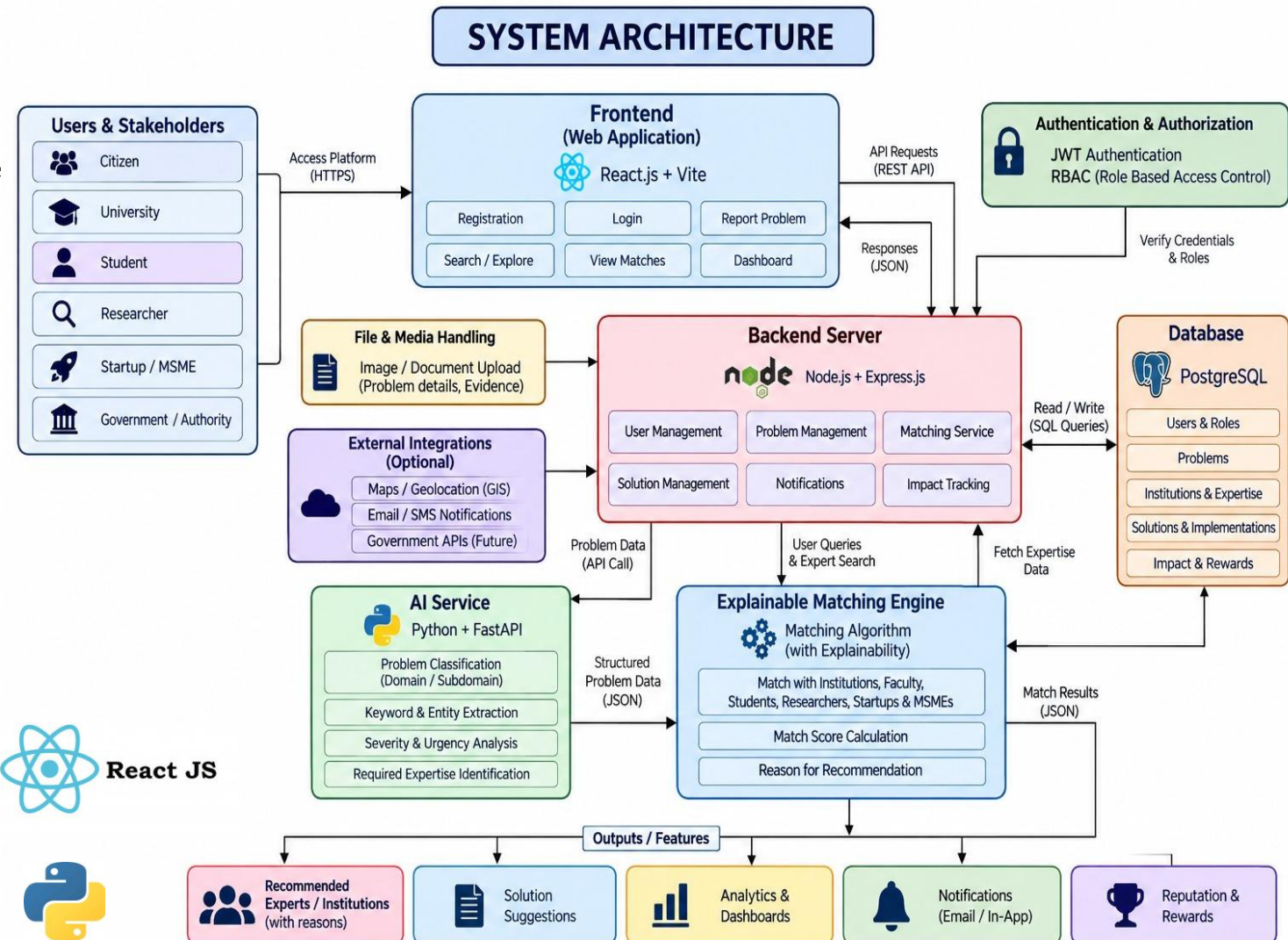
CivicSync does not stop at finding a problem or an expert — it creates a traceable path from
Problem → **Solution** → **Implementation** → **Verified Impact**

TECHNOLOGY STACK

- ❖ Frontend: HTML, Tailwind CSS, JavaScript, React.js
- ❖ Backend: Node.js, Express.js, REST APIs, bcrypt.
- ❖ RBAC(Role Based Access Control)
- ❖ Database: PostgreSQL
- ❖ Authentication: JWT Authentication
- ❖ AI/Intelligence: Python, FastAPI, Rule based NLP, Pattern Analysis, LLMs, Embeddings



SYSTEM ARCHITECTURE



FEATURES

- 1. Citizen Problem Reporting :**
Citizens can submit real-world societal problems with location, category and affected population.
- 2. AI Challenge Intelligence :**
AI identifies domain, subdomain, severity, urgency, keywords and required expertise from the problem.
- 3. Explainable Expertise Matching :**
Matches problems with relevant institutions, faculty, students, researchers, startups and MSMEs with match scores and reasons.
- 4. Duplicate & Similar Problem Detection :**
Identifies potentially duplicate or related problems to reduce repeated efforts.
- 5. Implementation & Pilot Tracking :**
Tracks milestones, progress, evidence and blockers during solution implementation.
- 6. Impact Measurement :**
Measures baseline vs actual outcomes, citizen feedback and verified impact.
- 7. Reputation, Rankings & Rewards :**
Rewards verified meaningful contributions through reputation points, badges and rankings.

Technical Feasibility

- Modular stack: React + Node.js + FastAPI + PostgreSQL — UI, workflow, DB, and AI cleanly separated
- MVP-ready AI: rule-based NLP + pattern matching — no GPU or massive datasets needed

Operational Feasibility

- Human-in-the-loop: AI classifies & ranks; authorities validate before action
- Multi-role RBAC: tailored dashboards for Citizens, Students, Universities, MSMEs, Govt

Data Feasibility

- Cold-start ready: bootstrapped with curated capability profiles for institutions, experts, startups
- Built for growth: structured schema, ready for future vector/LLM integration

Scalability & Sustainability

- Phased rollout: District → State → National deployment
- Plug-and-play growth: new civic domains, AI models, GIS layers, and external registries

VIABILITY FORMULA Buildable Now → Low-Cost MVP → Human-Controlled → Modular Ecosystem → Scalable Architecture

HONEST LIMITATION MVP demonstrates rule-based AI with curated capability profiles — semantic vector search, fine-tuned LLMs, and real-time civic-portal API integration are scoped as Phase 2 extensions.

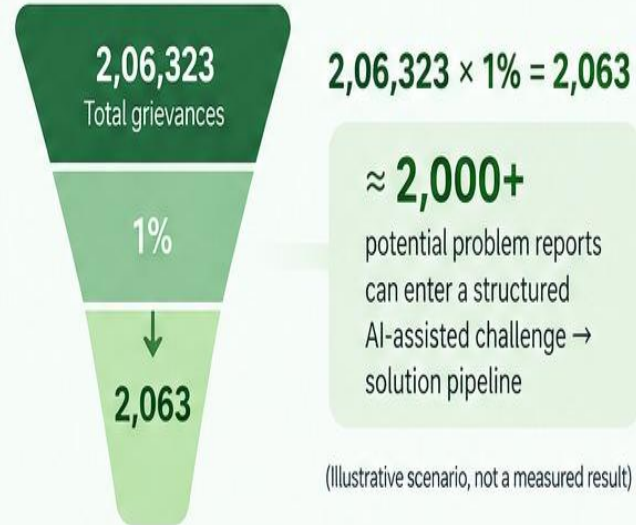
1. PROBLEM SCALE

CPGRAMS grievances received
(Central Ministries + States/UTs, June 2025)

2,06,323



Real citizen problems
across the country

2. EVEN 1% MAKES A DIFFERENCE**IMPACT**

 **80,000+**
Grievances / Month

- ❖ 75,914 received and 61,844 redressed in feb 2025.
- ❖ **Shows the need for better prioritization & follow-through.**

 **1.1 Cr+**
Students Engaged

- ❖ Atal Innovation Mission reports 1.1 crore + students engaged through Atal Tinkering Labs.
- ❖ **A huge pool of young problem-solvers exists.**

 **3,500+**
Startups Supported

- ❖ AIM reports 3,500+ startups supported through its incubation ecosystem.
- ❖ **Opportunity for a structured pathway from challenges to solution providers.**

 **66%**
City Portals Enable Complaints

- ❖ UN E-Government Survey 2024: 66% of surveyed city portals offer complaint/feedback submission.
- ❖ **Reporting is digital; the next step is solving and tracking.**

Benefits for Key Stakeholders

 **Citizens**

- Report local problems easily
- Track status and progress
- Get solutions from relevant experts

 **Students & Researchers**

- Find real-world problems.
- Gain practical experience.
- Work on meaningful projects.
- Recognition opportunities

 **Startups & MSMEs**

- Discover validated challenges
- Opportunities for pilot projects
- Access to academic expertise

 **Government / Authorities**

- Prioritize problems effectively.
- Access wider solution ecosystem.
- Track implementation and outcomes.
- Data-driven policy planning

EXPECTED IMPACT

- **Understand & Prioritize:** Scattered citizen reports are converted into structured, prioritized challenges, with related reports clustered to identify larger systemic problems and focus resources effectively.
- **Collaborate & Implement:** Academic solutions are connected with startups/MSMEs for implementation, while projects are tracked from **Baseline** → **Target** → **Actual** using evidence and verification.

**WHERE THE GAP EXISTS**

Existing platforms solve important parts, but not the complete journey.

Capability	Existing Platforms (CPGRAMS / MyGov / AIM-ARISE)	Our Platform (CivicSync)
 Citizen problem / grievance reporting	✓	✓
 Structured problem intelligence	~	✓
 Automated problem clustering	~	✓
 Explainable expertise matching	—	✓
 Implementation lifecycle tracking	~	✓

 = Core capability
  = Limited / Not core capability
  = Not a core capability

Expert Finding
Systems: A
Systematic Review

- ❖ **Evidence:** Research explores how relevant expertise can be identified for a particular problem or information need.
- ❖ **CivicSync:** CivicSync uses the requirements of a reported problem to identify relevant capabilities and potential contributors.
- ❖ **Source:** <https://www.mdpi.com/2076-3417/9/20/4250>

Automated Citizen
Request
Classification

- ❖ **Evidence:** Research shows that large volumes of unstructured citizen requests can benefit from automated classification.
- ❖ **CivicSync:** AI-based problem intelligence for domain, severity, urgency and required expertise extraction.
- ❖ **Source:** <https://www.sciencedirect.com/science/article/pii/S0306457320309055>

Citizen Complaint
Pattern Analysis

- ❖ **Evidence:** Research on citizen complaints identifies spatial and temporal patterns that can reveal recurring or concentrated issues.
- ❖ **CivicSync:** Related-problem clustering and creation of systemic/master challenges.
- ❖ **Source:** <https://www.sciencedirect.com/science/article/pii/S2226585624000037>

Government/Ecosystem**CPGRAMS**
Government
of India

- ❖ Centralized public grievance reporting, tracking, feed backend appeal.
- pgportal.gov.in

MyGov
Government
of India

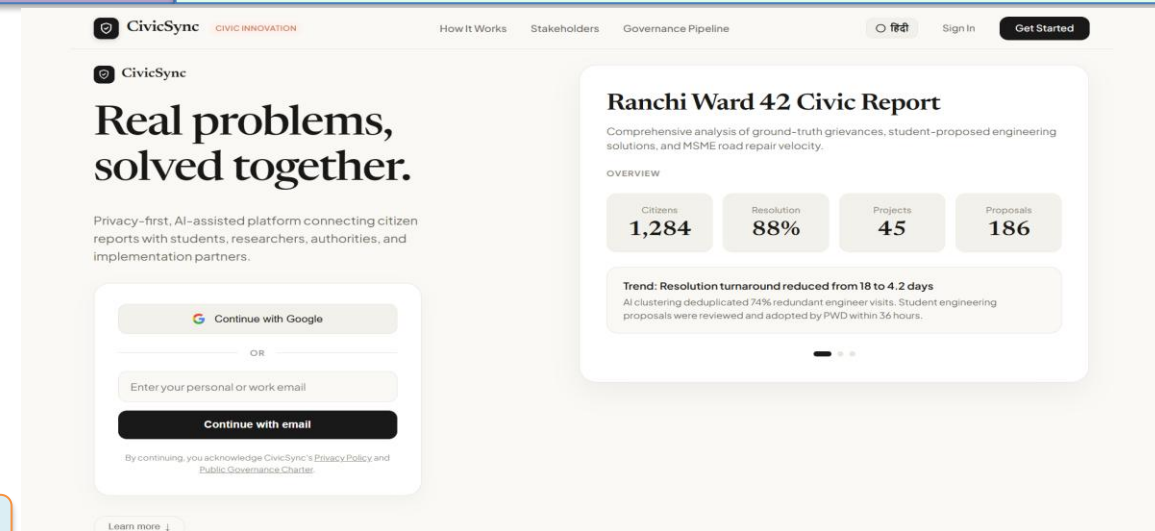
- ❖ Citizen engagement through tasks, discussions, polls, campaigns and submissions
- mygov.in

**Atal
Innovation
Mission
NITI-Aayog**

- ❖ Citizen engagement through tasks, discussions, polls, campaigns and submissions
- aim.gov.in

**Manthan —
Principal
Scientific
Adviser**

- ❖ Multi-stakeholder collaboration across government, academia & industry.
- psa.gov.in/manthan

Live Link:- <https://civicsync-frontend.onrender.com/>Git Hub Repo:- github.com/ayushcodes74/SIHPS26043